

Auspify Tasks — Project Setup & Execution Guide

Clean, minimal guide to running and evaluating the 4 full-stack Next.js applications

► Video Demonstration & Review Walkthrough

A recorded video walkthrough covering all four applications, authentication flows, interactive dashboards, and database operations without needing to start local servers:

<https://youtu.be/GSONik1QADQ?si=qM3gDQe1Q+tST0J2g>

1. SYSTEM REQUIREMENTS

Component	Supported Version	Notes
Node.js	v20.x or v22.x LTS	Minimum v18.18+ ; Active LTS 20 recommended
Package Manager	npm 10.x+ or pnpm 9.x+	Standard npm included with Node.js
Database	MongoDB Atlas or Local MongoDB 6.0+	Pre-configured Atlas credentials provided in each .env.local
Operating System	Linux, macOS, Windows (WSL2)	Standard POSIX shell or PowerShell

2. QUICK START (STANDARD 3-STEP PATTERN)

Each application in this monorepo is fully standalone and decoupled. To start any application:

```
# 1. Enter the application directory
cd <project_name>    # expense_tracker, job_portal_application, etc.

# 2. Install dependencies
npm install

# 3. Start local development server
npm run dev          # Serves at http://localhost:3000
```

3. SECURITY & ENGINEERING STANDARDS

All four applications adhere to the same production-grade conventions:

- **Database Integrity:** Real MongoDB queries and mutations across all views; zero mock data.
- **Authentication:** Argon2id password hashing + dual-cookie JWT architecture with refresh token rotation.
- **Input Validation:** Strict Zod schema parsing on all request bodies, headers, and query parameters.
- **Responsive Design:** Tested across mobile (360px+), tablet (768px+), and desktop displays.

4. APPLICATION SETUP & PRELOADED ACCOUNTS (PART 1)

1. Expense Tracker & Wealth Intelligence (expense_tracker/)

Default Port: 3000

Personal finance platform featuring integer minor-unit math, MongoDB aggregation pipelines, and dynamic budgets.

```
cd expense_tracker
npm install
npm run dev                # Serves at http://localhost:3000
```

Account Type	Email	Password	Preloaded Data
Interactive Demo	demo@auspify.com	DemoUser1234!	100+ transactions, active budgets, analytics

2. Job Portal Application (job_portal_application/)

Default Port: 3000 / 3001

Paperfolio Neo-Brutalist hiring platform with profile snapshots, candidate pipeline tracking, and admin moderation.

```
cd job_portal_application
npm install
npm run db:index          # Sync database compound indexes
npm run db:seed           # Seed categories, companies, jobs, and accounts
npm run dev               # Serves at http://localhost:3000
```

Role	Email	Password	Preloaded Context
Job Seeker	seeker@demo.local	DemoPassword123!	Complete profile, submitted applications, saved jobs
Employer	employer@demo.local	DemoPassword123!	Company "Vortex Systems", active listings, candidate funnel
Administrator	admin@demo.local	DemoPassword123!	Platform moderation, audit logging, user governance

Database Note: Live MongoDB Atlas connection strings are already configured in each project's `.env.local`. You can start the dev servers immediately without creating your own database instance.

5. APPLICATION SETUP & PRELOADED ACCOUNTS (PART 2)

3. Enterprise Learning Management System (learning_management_system/)

Default Port: 3000 / 3002

E-learning platform with timed quizzes, curriculum video/markdown player, grading queue, and verifiable certificates.

```
cd learning_management_system
npm install
npm run db:seed          # Seed courses, modules, and multi-role accounts
npm run dev              # Serves at http://localhost:3000
```

Role	Email	Password	Preloaded Context
Student	student@lms.local	StudentDemo123!	Active enrollments, quiz evaluations, certificates
Instructor	instructor@lms.local	InstructorDemo123!	Curriculum studio, assignment authoring, grading queue
Administrator	admin@lms.local	SuperSecretAdminPassword123!	Course moderation, audit trail, user role management

4. Student Management System (student_management_system/)

Default Port: 3000 / 3003

Academic administration tool with PII protection, cohort capacity tracking, and typed deletion safeguards.

```
cd student_management_system
npm install
npm run db:index          # Sync database indexes
npm run db:seed           # Seed cohorts, students, and staff accounts
npm run dev               # Serves at http://localhost:3000
```

Role	Email	Password	Preloaded Context
Administrator	admin@studentms.internal	AdminStrongPass123!	Institutional metrics, staff provisioning, audit logs
Academic Staff	staff@studentms.internal	StaffTempPass123!	Restricted staff role, triggers first-login password gate

6. RUNNING APPLICATIONS SIMULTANEOUSLY & TROUBLESHOOTING

Running on Custom Ports:

To run multiple apps concurrently, prefix with `PORT`:

```
# Terminal 1: Expense Tracker
cd expense_tracker && PORT=3000 npm run dev

# Terminal 2: Job Portal
cd job_portal_application && PORT=3001 npm run dev

# Terminal 3: LMS
cd learning_management_system && PORT=3002 npm run dev

# Terminal 4: Student MS
cd student_management_system && PORT=3003 npm run dev
```

Troubleshooting Notes:

- **Atlas Whitelist:** Ensure your IP is allowed in MongoDB Atlas Network Access if using a custom database.
- **Automated Tests:** Run `npm test` inside any project to run Vitest unit and integration suites against in-memory databases.
- **Node Version:** Verify version with `node -v` (v20+ recommended).